NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-26/3C
1 June 1976
SUPERSEDING
MIL-R-26/3B
11 May 1973

PERFORMANCE SPECIFICATION

RESISTORS, FIXED, WIRE-WOUND (POWER TYPE),

STYLES RW29, RW301/, RW31, RW321/, RW33, RW341/, RW35, RW361/,

RW37, RW38, RW391/, AND RW47

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements for styles RW29, RW301/, RW31, RW321/, RW33, RW341/, RW35, RW361/, RW37, RW38, RW391/, and RW47 resistors.

1.2 Classification.

1.2.1 Type designation. The type designation shall be in the following form:

RW29	V	100
Style (1.2.1.1)	Characteristic (1.2.1.2)	Resistance (1.2.1.3)

1.2.1.1 Style. The style is identified by the two-letter symbol "RW" followed by a two-digit number; the letters identify power type, wire-wound, fixed resistors, and the number identifies the size, wattage rating, and construction of the resistors.

* 1.2.1.2 Characteristic. The characteristic is identified by a single letter that identifies the method of winding, maximum continuous operating temperature (surface hot spot), the minimum insulation resistance value at the end of the moisture resistance test, and the resistance temperature characteristic in accordance with table I.

TABLE I. Characteristic.

Symbol	Method of winding	Maximum continuous operating temperature 1/	Minimum insulation resistance at end of moisture resistance test	Resistance-temperature characteristic (percent per degree C)
		°C	Megohms	
V - - -	Inductive	350	100	±.040 for resistance values below 20 ohms. ±.026 for resistance values of 20 ohms and above.
N - - -	Noninductive			

1/ This temperature is also the maximum permissible hot-spot surface temperature.

1/ Not to be used for new design. For replacement purposes only.

FSC 5905

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1.2.1.3 Resistance. The nominal resistance expressed in ohms is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeroes to follow. Where fractional values of an ohm are required, the letter "R" shall be substituted for one of the significant digits to represent the decimal point. When the letter "R" is used, succeeding digits of the group represent significant figures as shown in the following example:

R10 = 0.1 ohm
1R0 = 1.0 ohm

1.2.1.4 Center-tapped resistors. The letter "T" when added at the end of the type designation shall denote that the resistor is center-tapped.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-R-26 - Resistors, Fixed, Wire-Wound (Power Type), General Specification for.

STANDARDS

MILITARY

MS90178 - Standard 24-Value Series Decade for Electronic Components of 5-, 10-, and 20-Percent Tolerances.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-26, and as specified herein.

3.2 Design and construction. Resistors shall be of the design, construction, and physical dimensions as specified on figure 1, and as specified herein.

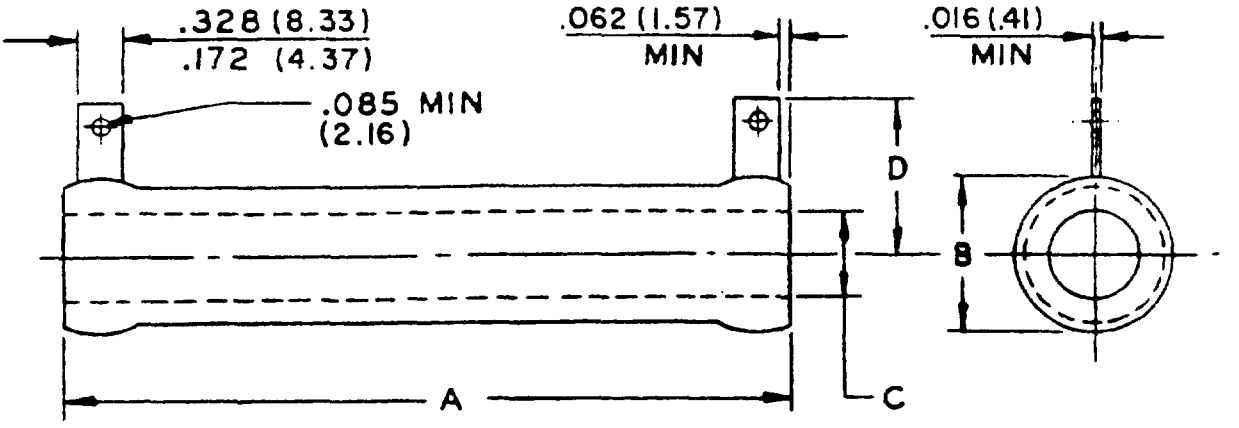
* 3.3 Power rating. The power rating shall be as specified in table II, based on full load operation at an ambient temperature of 25°C.

TABLE II. Power rating.

Resistor style	Power rating in watts		NATO style 1/ (per NEPR No. 5)	NATO power rating at 70°C (watts)
	Characteristic			
	N	V		
RW29 - - - - - -	11	11	NRW01	9.5
RW30 - - - - - -	---	11	---	---
RW31 - - - - - -	14	14	NRW02	12
RW32 - - - - - -	---	17	---	---
RW33 2/ - - - - -	26	26	NRW03	22.5
RW34 2/ - - - - -	---	43	---	---
RW35 2/ - - - - -	55	55	NRW04	47.5
RW36 2/ - - - - -	---	78	---	---
RW37 2/ - - - - -	113	113	NRW05	97.5
RW38 2/ - - - - -	159	159	NRW06	137
RW39 2/ - - - - -	---	240	---	---
RW47 2/ - - - - -	210	210	NRW07	181

1/ Not applicable to characteristic N.
2/ The power rating of center-tapped resistors shall be 90 percent of those shown.

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	Style	A .062 (1.57)	B, Max	C	D±.125 (3.18)
	RW29	1.750 (44.45)	.500 (12.70)	.172 Min (4.37)	.625 (15.88)
*	RW30 1/	1.000 (25.40)	.594 (15.09)	.187 Min (4.75)	.625 (15.88)
	RW31	1.500 (38.10)	.594 (15.09)	.312+.016,-.125 (7.92 +.41,-3.18)	.625 (15.88)
*	RW32 1/	2.000 (50.80)	.594 (15.09)	.312+.016,-.125 (7.92 +.41,-3.18)	.625 (15.88)
	RW33	3.000 (76.20)	.594 (15.09)	.187 Min (4.75)	.625 (15.88)
	RW34 1/	3.000 (76.20)	.906 (32.01)	.516+.046,-.047 (13.11+1.17,-1.19)	.812 (20.62)
	RW35	4.000 (101.60)	.906 (23.01)	.500+.062,-.031 (12.70+1.57, -.79)	.812 (20.62)
*	RW36 1/	4.000 (101.60)	1.312 (33.32)	.750+.156,-.047 (19.05+3.96,-1.19)	1.219 (30.96)
	RW37	6.000 (152.40)	1.312 (33.32)	.703 Min (17.86)	1.219 (30.96)
	RW38	8.000 (203.20)	1.312 (33.32)	.750+.156,-.047 (19.05+3.96,-1.19)	1.219 (30.96)
	RW39 1/	12.000 (304.80)	1.312 (33.32)	.750+.156,-.047 (19.05+3.96,-1.19)	1.219 (30.96)
	RW47	10.500 (266.70)	1.312 (33.32)	.703 Min (17.86)	1.219 (30.96)

* 1/ Not to be used for new design; for replacement purposes only.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 1. Styles RW29, RW30, RW31, RW32, RW33, RW34, RW35, RW36, RW37, RW38, RW39, and RW47.

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3.4 Resistance. The minimum and maximum nominal resistance values shall be as specified in table III.

TABLE III. Minimum and maximum nominal resistance value.

Resistor style	Resistance value		
	Minimum	Maximum	
		Characteristic V	Characteristic N
	ohm	ohms 1/	ohms 1/
RW29	0.10	5,600	2,700
RW30 2/	0.10	2,700	---
RW31	0.10	6,800	3,300
RW32 2/	0.10	10,000	---
RW33	0.10	18,000	8,100
RW34 2/	0.10	30,000	---
RW35	0.10	43,000	20,000
RW36 2/	0.10	56,000	---
RW37	0.10	91,000	43,000
RW38	0.10	0.15 megohm	75,000
RW39 2/	0.10	.20 megohm	---
RW47	0.10	.18 megohm	81,000

1/ Based on the use of 0.00175-inch nominal diameter wire.
2/ Not to be used for new design. For replacement purposes only.

3.4.1 Standard resistance values. For standard resistance values, see MS90178.

3.5 Resistance tolerance. These resistors have a resistance range of 0.10 ohm to 0.2 megohm with the following resistance tolerances:

(a) Resistors of less than 1 ohm in resistance value shall have a tolerance of ±10 percent of the nominal resistance value. If center-tapped, the resistance tolerance between tap and terminal shall be ±10 percent.
(b) Resistors of 1 ohm and over in resistance value shall have a tolerance of ±5 percent of the nominal resistance value. If center-tapped, the resistance tolerance between tap and terminal shall be ±10 percent.

3.6 Center-tapped resistors. Center-tapped resistors are available in styles RW33, RW34, RW35, RW36, RW37, RW38, RW39, and RW47.

3.7 Thermal shock. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.8 Short-time overload. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.9 High-temperature exposure. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.10 Moisture resistance. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.11 Low-temperature storage. The resistors shall not change in resistance in excess of ±(2 percent +.05 ohm).

3.12 Life. The resistors shall not change in resistance in excess of ±(3 percent +.05 ohm).

3.13 Solderability. Solderability is applicable to this specification.

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3.14 Terminal strength. The resistors shall not change in excess of ±(1 percent + .05 ohm).

3.15 Dielectric withstanding voltage (barometric pressure, reduced, not applicable). The resistors shall not change in resistance in excess of ±(.1 percent + .05 ohm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-26, and as specified herein.

4.2 Short-time overload. The maximum voltages shall be in accordance with table IV.

TABLE IV. Maximum voltages for short-time overload.

Resistor style	Voltage (volts, maximum)	Resistor style	Voltage (volts, maximum)
RW29 - - - - - -	1,200	RW35 - - - - - -	3,000
RW30 - - - - - -	450	RW36 - - - - - -	3,000
RW31 - - - - - -	1,000	RW37 - - - - - -	5,000
RW32 - - - - - -	1,400	RW38 - - - - - -	7,000
RW33 - - - - - -	2,400	RW39 - - - - - -	12,000
RW34 - - - - - -	2,400	RW47 - - - - - -	10,000

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-R-26.

6. NOTES

6.1 Notes. In addition to the notes specified herein, the notes specified in MIL-R-26 are applicable to this specification.

6.2 Supplementary insulation. Where potential to ground is over 500 volts, supplementary insulation should be provided.

6.3 Mounting. These resistors should not be mounted by their terminals.

6.3.1 Bracket assemblies. When required, bracket assemblies (mounting hardware) are available for these resistors under MS75009, Bracket Assembly, Resistor (Power type).

6.4 Derating. Resistors shall be derated, when necessary, in accordance with figure 2.

6.5 Supersession data. This specification supersedes MIL-R-27232(USAF), dated 8 September 1959, "Resistors, Fixed, Wirewound (Power Type), Styles RW34 and RW39."

6.6 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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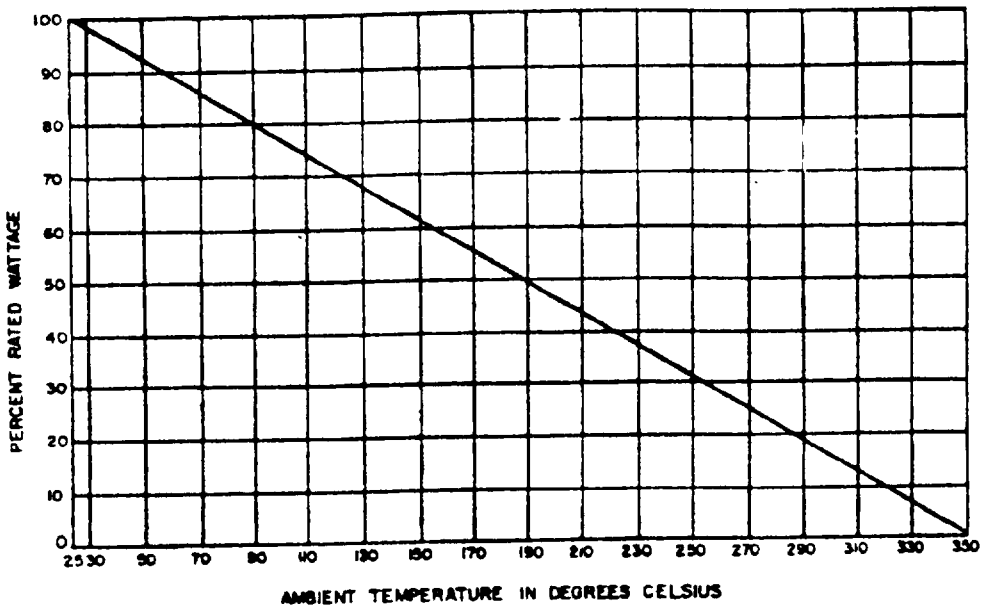


FIGURE 2. Derating curve for high ambient temperatures.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, MU, EL
Navy - EC, AS, OS
Air Force - 11, 17, 80, 85
DSA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - EL

Agent:
DSA - ES

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